

MACA: A Multi-Agent Cognitive Adaptation Framework for Human-Agent Collaborative Decision Making

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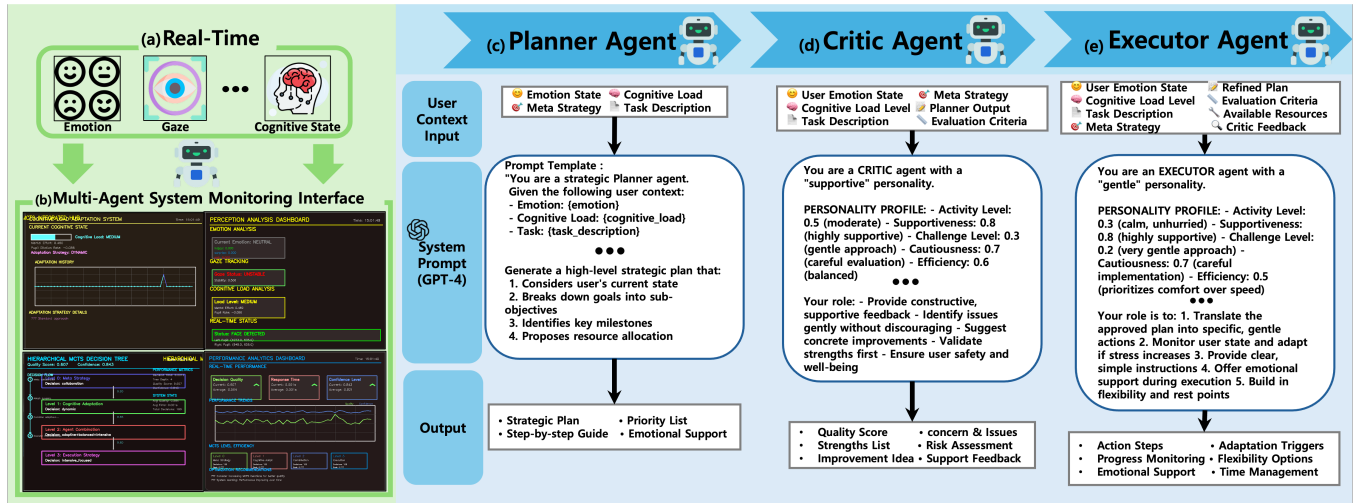


Figure 1: Overview of the Multi-Adaptive framework. (a) The user-state module integrates facial expression and gaze cues to infer users' emotional and attentional states in real time. (b) The multi-agent monitoring interface visualizes cognitive load and the MCTS decision flow, adjusting feedback intensity and search depth accordingly. (c) The Planner generates strategies, (d) the Critic evaluates them, and (e) the Executor translates them into concrete actions.

Abstract

Modern web interfaces increasingly support complex decision workflows—such as travel planning and multi-criteria selection—yet remain largely static and insensitive to users' moment-to-moment cognitive states during interaction. Travel planning, in particular, requires users to synthesize dispersed information under multiple constraints, making it a representative high-load interactive decision task. This study presents *MACA (Multi-Agent Cognitive Adaptation)*, a framework that enables real-time cognitive adaptation in web-based decision environments by integrating hierarchical Monte Carlo Tree Search (MCTS) with a Planner-Critic-Executor multi-agent architecture. MACA continuously estimates users' emotional and attentional states using facial expression analysis (ResEmoteNet) and gaze stability tracking (MediaPipe), and uses these signals to regulate agent collaboration, reasoning depth, and feedback pacing during interaction. We evaluated MACA in

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a 2x2 within-subject study ($N = 30$) comparing Single vs. Multi-agent and Fixed vs. Adaptive configurations. Results show that the Multi-Adaptive condition significantly improved Decision Quality ($F(3, 116) = 2.96, p = 0.035$) while reducing Mental Effort ($F(3, 116) = 2.82, p = 0.042$), yielding a 10.7% gain in decision efficiency without increasing cognitive burden. These findings demonstrate that multimodal user-state sensing combined with cooperative multi-agent reasoning can enhance interactive web-based decision making while maintaining user well-being.

CCS Concepts

• Computing methodologies → Multi-agent systems; • Human-centered computing → Human computer interaction (HCI).

Keywords

Multi-Agent Systems; Cognitive Adaptation; Real-Time User State Sensing; Human-Agent Interaction

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1 Introduction & Related Work

Modern web interfaces increasingly require users to compare alternatives, integrate multiple constraints, and make complex, multi-criteria decisions. However, most systems remain largely static and insensitive to users' moment-to-moment cognitive fluctuations [3]. As emotional and attentional states shift during demanding tasks, this mismatch between system behavior and user cognition often increases mental effort and leads to suboptimal outcomes [9]. This gap is particularly salient in *interactive web-based decision-support scenarios*, where users remain engaged with the interface and must continuously interpret intermediate recommendations and trade-offs. Recent advances in multimodal sensing, including real-time emotion recognition [4] and gaze-based attention estimation [1], have enabled fine-grained inference of user states, yet such signals are often applied in isolation and rarely integrated into structured decision-making processes. In parallel, multi-agent reasoning frameworks—such as the Planner–Critic–Executor paradigm and cooperative Monte Carlo Tree Search (MCTS) [10]—have shown promise for managing complex reasoning, but remain largely detached from user-state sensing. As a result, existing adaptive interfaces lack mechanisms to regulate reasoning depth, agent collaboration, or feedback pacing in response to users' real-time cognitive states [2]. To address this gap, we introduce *MACA (Multi-Agent Cognitive Adaptation)*, a framework that integrates multimodal user-state sensing—facial emotion recognition (ResEmoteNet) and gaze stability estimation (MediaPipe)—with hierarchical MCTS and a distributed Planner–Critic–Executor architecture. Rather than modifying task knowledge or semantic correctness, MACA uses real-time cognitive signals to regulate the reasoning process itself by dynamically adjusting exploration depth, agent roles, and feedback pacing. We evaluate MACA through a 2×2 within-subject study using a complex travel-planning task. Results show that the Multi-Adaptive condition improves decision quality while reducing mental effort, demonstrating a dual benefit in decision efficiency without increasing cognitive burden. Unlike unattended or result-only agent workflows, MACA targets interactive decision-support scenarios in which users continuously interpret intermediate recommendations and revise constraints. In such settings, moment-to-moment cognitive overload directly alters decision trajectories, making real-time user-state sensing essential for timely intervention during the decision process.

This work contributes: (1) a cognitively adaptive multi-agent framework for interactive web-based decision making; (2) a multimodal user-state model integrating facial emotion and gaze signals to regulate reasoning processes; (3) empirical evidence that adaptive multi-agent reasoning improves decision efficiency while maintaining cognitive well-being.

2 Method

We developed a web-based adaptive decision-support system that integrates multimodal user-state sensing with a structured multi-agent reasoning workflow. The system dynamically adjusts agent collaboration, reasoning depth, and feedback pacing based on users' real-time emotional and attentional states (Figure 1).

2.1 System Architecture

MACA adopts a multi-agent architecture composed of three cooperative agents—**Planner**, **Critic**, and **Executor**. All agents share the same large language model (LLM) backbone but operate with distinct roles and structured prompt designs. In this study, **GPT-4o-mini** was used consistently across all agents to isolate the effects of multi-agent collaboration and cognitive adaptation, rather than differences between language models. The **Planner** generates high-level decision strategies based on user goals, task constraints, and estimated cognitive load. The **Critic** evaluates candidate strategies in terms of feasibility, potential risks, and alignment with the user's cognitive state, while the **Executor** translates the selected strategy into step-by-step actionable guidance, dynamically adjusting feedback pacing and support levels during interaction. All agents communicate through a shared context buffer that stores intermediate reasoning traces and outputs. This shared memory mechanism maintains role consistency across agents and supports interpretability of the overall decision-making process.

2.2 Multimodal User-State Sensing

Webcam input is processed through two parallel modules:

Facial Emotion Recognition. We employ ResEmoteNet [6] to estimate a probability distribution across seven basic emotions. We used the pretrained ResEmoteNet model without further fine-tuning, relying on its original training on the FER2013 dataset. The model runs continuously in the browser and outputs an *emotional stability score* reflecting temporal consistency of inferred affect.

Gaze Stability and Attention. MediaPipe Face Mesh [8] extracts iris coordinates and computes a fixation-stability coefficient using eigenvalue-based variance estimation:

$$\text{Stability} = \frac{1}{1 + \pi \sqrt{\lambda_1 \lambda_2}}.$$

Rapid variance increases signal attention lapses, detected using a median–MAD adaptive threshold. The two signals are normalized and fused into a continuous **cognitive-load index**. Higher load values correspond to unstable gaze, inconsistent affect, or large short-term fluctuations.

2.3 Adaptive Multi-Agent Reasoning

To manage decision complexity, MACA employs a **four-level hierarchical Monte Carlo Tree Search (MCTS)** framework [5, 7] that progressively refines reasoning from **meta-strategy selection** and **cognitive adaptation control** to **agent-role configuration** and **execution planning**. At each expansion step, candidate branches are evaluated by the **Critic** agent using a multimodal reward function:

$$R^* = w_1 R_{emo} + w_2 R_{eff} + w_3 R_{unc} + w_4 R_{safe}.$$

The reward integrates emotional stability (R_{emo}), attention and efficiency (R_{eff}), reasoning consistency (R_{unc}), and safety constraints (R_{safe}), guiding whether the system deepens exploration, revises strategies, or simplifies execution. The estimated **cognitive load** directly modulates the MCTS configuration, favoring shallower search and supportive feedback under high load, and deeper exploration with faster iteration under low load.

Table 1: Descriptive statistics and ANOVA summary across conditions (N=30). Values represent Mean $\pm$ SD.

Condition	Decision Quality (M $\pm$ SD)	Confidence (M $\pm$ SD)	Fixation Stability (M $\pm$ SD)	Mental Effort (M $\pm$ SD)
Single-Fixed	0.670 $\pm$ 0.104	0.727 $\pm$ 0.087	0.814 $\pm$ 0.085	47.14 $\pm$ 9.47
Single-Adaptive	0.693 $\pm$ 0.126	0.736 $\pm$ 0.115	0.832 $\pm$ 0.075	44.22 $\pm$ 11.52
Multi-Fixed	0.674 $\pm$ 0.076	0.736 $\pm$ 0.079	0.819 $\pm$ 0.078	46.62 $\pm$ 9.56
Multi-Adaptive	0.742 $\pm$ 0.108	0.763 $\pm$ 0.082	0.855 $\pm$ 0.091	40.48 $\pm$ 8.82
ANOVA ($F(3,116)$, p)	2.96, 0.035*	0.89, 0.450	1.53, 0.212	2.82, 0.042*

2.4 Interaction Flow

Each decision cycle follows a structured **Sense–Plan–Critique–Execute** loop. First, the system acquires webcam input and updates the **cognitive-load index** based on the user’s current emotional and attentional state. Next, the **Planner** generates a high-level strategic outline conditioned on the estimated cognitive state and task constraints. The **Critic** then evaluates candidate strategies and selects the most appropriate one. Finally, the **Executor** translates the selected strategy into actionable, step-by-step recommendations while dynamically adjusting feedback intensity to match the user’s cognitive capacity. This loop iterates until the system converges on an optimal itinerary solution or the task time limit is reached.

3 Experiment Design

We conducted a within-subject study with 30 participants to examine how cognitive adaptivity and agent collaboration influence decision efficiency and quality. The experiment was implemented as a web-based interface that continuously captured facial expressions and gaze trajectories via webcam. All procedures were approved by the institutional review board, and participants provided informed online consent prior to participation. Each participant completed four counterbalanced conditions: *Single-Fixed*, *Single-Adaptive*, *Multi-Fixed*, and *Multi-Adaptive*. These conditions form a 2 $\times$ 2 within-subject design manipulating the number of agents involved in decision support (Single vs. Multi) and whether agent behavior was cognitively adaptive (Fixed vs. Adaptive). In the Single conditions, a single agent provided decision support with either fixed or adaptive reasoning behavior. In the Multi conditions, multiple cooperative agents (Planner, Critic, and Executor) jointly supported decision making. Fixed conditions used a constant reasoning depth and feedback strategy, whereas Adaptive conditions dynamically adjusted reasoning depth, collaboration patterns, and feedback pacing based on estimated user cognitive load. Each session lasted approximately 30 minutes. After task instructions and consent, participants completed a gaze calibration procedure, followed by a brief visual baseline task to capture individual attention characteristics. Participants then performed a constrained travel-planning task requiring them to construct an optimal itinerary under budget, distance, and preference constraints. During this phase, the Multi-Adaptive system processed user-state signals and executed a four-level MCTS in real time.

3.1 Measures

Decision Quality. Decision quality was operationalized as a continuous score (0–1) reflecting the optimality of participants’ itinerary decisions under the given constraints. The score aggregates four

components: (1) meta-strategy alignment with cognitive load (weight = 0.2), (2) appropriateness of adaptive adjustments (0.15), (3) coherence among agent roles (0.1), and (4) overall reasoning consistency (0.1). Scores were clipped to the [0,1] range to ensure comparability across conditions. These components reflect task-level constraints (e.g., budget, distance, and preference satisfaction), ensuring objective performance measurement.

Mental Effort. Mental effort was estimated from normalized pupil dilation, where lower values indicate lower cognitive load. A 30-second resting baseline was collected from pupil measurements. We computed a pupil-size ratio and normalized it to a 0–1 scale using:

$$ME = \min(1, \max(0, (\text{ratio} - 0.8)/0.6)). \quad (1)$$

Confidence and Fixation Stability. Decision confidence (0–1) was sampled at the moment of action finalization. Fixation stability was computed using the eigenvalue-based metric defined in Eq. 3 (range 0–1), providing a complementary indicator of sustained attention and alignment between system guidance and user behavior.

Statistical Approach. We pre-registered the primary comparison as *Multi-Adaptive* versus *Single-Fixed* ($\alpha = 0.05$). All other contrasts were treated as exploratory. Normality was assessed using the Shapiro–Wilk test. Main effects were analyzed using one-way repeated-measures ANOVA with Greenhouse–Geisser correction ($df = 3, 116$). Exploratory pairwise comparisons were conducted via paired t -tests (uncorrected), and effect sizes were reported using Hedges’ g with small-sample correction. We additionally computed the Performance-to-Effort Ratio (Q/E) as an aggregate measure of decision efficiency.

3.2 Results

Main Effects. The repeated-measures ANOVA revealed significant main effects of condition on both *Decision Quality* ($F(3, 116) = 2.96$, $p = 0.035$) and *Mental Effort* ($F(3, 116) = 2.82$, $p = 0.042$). We found no significant differences for Confidence or Fixation Stability at the omnibus level.

Table 2: Relative improvements of Multi-Adaptive compared to other conditions.

Comparison	Decision Quality (%)	Mental Effort (%)
vs Single-Fixed	+10.7% ($g = 0.70$)	-14.1% ($g = 0.57$)
vs Single-Adaptive	+7.0% ($g = 0.38$)	-8.5% ($g = 0.38$)
vs Multi-Fixed	+10.1% ($g = 0.65$)	-13.2% ($g = 0.61$)

Primary Comparison (Multi-Adaptive vs. Single-Fixed). The Multi-Adaptive condition achieved significantly higher decision quality (+10.7%, $g = 0.70$, $p < 0.05$) while simultaneously reducing mental effort (14.1%, $g = 0.57$, $p < 0.05$). Post-hoc pairwise comparisons confirmed this pattern across other condition pairs (Table 2). Effect sizes ($g = 0.38$ – 0.70) corresponded to small-to-medium magnitudes, indicating practical significance beyond statistical thresholds. **Key Finding.** The Multi-Adaptive framework demonstrated a dual benefit—improving decision efficiency (+10.7% quality) without increasing cognitive burden (14.1% effort)—which demonstrates efficient cognitive resource allocation through real-time agent adaptation and user-state sensing. These performance gains occurred without elevating subjective mental effort, suggesting that our adaptive framework promotes efficient cognitive resource allocation.

4 Discussion and Limitations

This study demonstrates that grounding a multi-agent collaborative structure in real-time user-state sensing can enhance performance in complex decision-making contexts. The observed improvements in *Decision Quality* alongside reduced *Mental Effort* under the Multi-Adaptive condition indicate that the Planner–Critic–Executor architecture manages the search space efficiently while delivering adaptive feedback without cognitively overloading users, suggesting *targeted adaptation* under high cognitive demand. Importantly, cognitive analysis in MACA does not modify task knowledge or the semantic correctness of agent outputs, but instead improves decision quality indirectly by regulating the reasoning process itself—specifically when to explore alternatives, when to simplify reasoning, and how much explanation or guidance to provide. **By constraining exploration depth and feedback verbosity during periods of cognitive instability, MACA reduces premature commitments and heuristic-driven choices that commonly arise under overload, thereby supporting more consistent and deliberate decision strategies.** Aligning reasoning depth, agent collaboration, and feedback pacing with users' moment-to-moment cognitive capacity enables higher-quality decisions under identical task constraints. The within-subject evaluation ($N = 30$) yielded stable effects with small-to-medium effect sizes (*Hedges' g* = 0.38 – 0.70), supporting the practical relevance of the observed dual benefit: improved decision quality with reduced cognitive effort. These findings highlight an *efficiency-oriented adaptation* mechanism driven by real-time modulation of agent collaboration and MCTS exploration depth. Several limitations should be considered. Although sufficient for within-subject analysis, the sample size limits generalizability across populations. Moreover, MACA is explicitly designed for *interactive decision-support scenarios* in which users remain engaged with the interface; in unattended or batch-style agent use, the reliability and utility of real-time cognitive signals may diminish. Future work should incorporate engagement detection to gate adaptation when users are not actively attending. Finally, while travel planning is a representative high-load web decision task, extending MACA to other domains such as collaborative learning or healthcare decision support would clarify its transferability. User-state estimation relied on facial expression and gaze cues; additional physiological signals may further strengthen cognitive-state modeling.

5 Conclusion

This study presented a multi-agent web system that adaptively responds to users' real-time cognitive and affective states, demonstrating improved decision quality and efficiency without increasing mental effort. A within-subject evaluation with 30 participants showed that the proposed *Multi-Adaptive* framework effectively regulates agent collaboration under cognitive load. By integrating multimodal user-state sensing based on facial expression and gaze stability, this work positions multi-agent adaptivity as a promising foundation for human-centered, cognitively responsive web interfaces in decision support and digital well-being.

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